

Current State Sequence Diagrams

Complete System Architecture & Component Interaction Flows

STATUS: 100% IMPLEMENTED

Date: October 3, 2026

Scope: 9 Core Workflows (Auth, POS, Memberships, Discounts, Management)

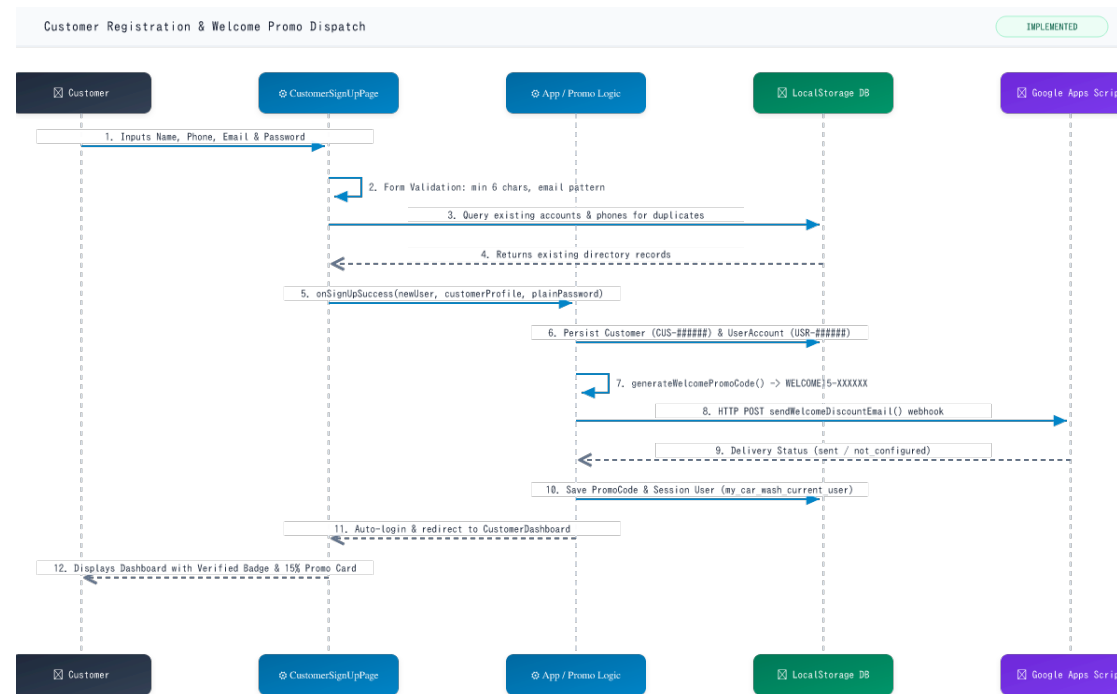
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Workflow 1: Customer Registration & Welcome Promo Dispatch

STATUS: IMPLEMENTED

Purpose: Allows a new guest customer to register an account on the public portal, validates uniqueness, generates customer CRM and user account credentials, issues a 15% First-Signup welcome coupon, and triggers an email dispatch webhook.



Workflow Summary & Key Components:

Execution Flow:

- Customer fills first name, last name, phone, email, password, and confirm-password fields on the public signup page.
- The client component validates field presence, matching passwords, and normalized phone digits.
- A duplicate lookup queries localStorage to ensure the email and phone do not already belong to an active customer. If duplicate, a direct "Sign in here" link appears.
- Upon submission, App.tsx generates a sequential customer profile (CUS-#####) and salted user account hash.

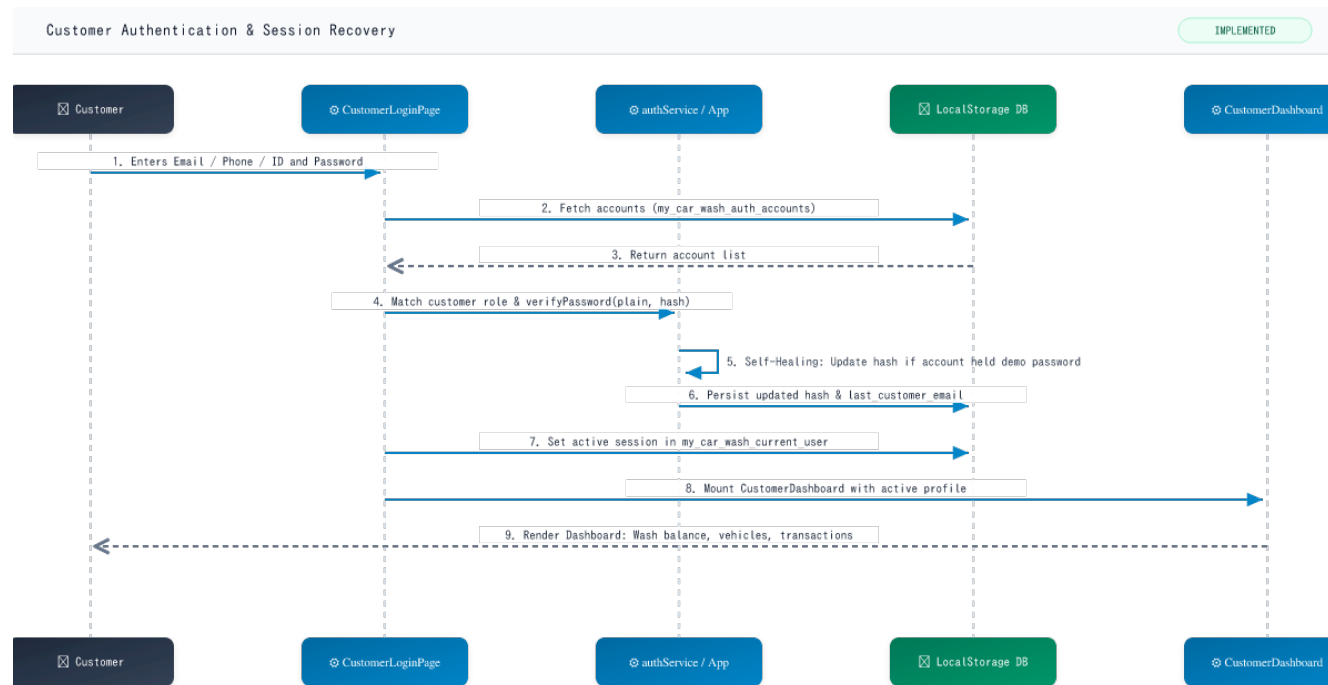
Enforced Business Rules & Error Handling:

- ' Password must be at least 6 characters.
- ' Phone numbers are normalized via non-digit regex to prevent duplicate entries.
- ' Welcome discount is strictly set to 15% with a 3-month expiration window.
- & Duplicate email/phone -> Inline alert with error message and pre-filled sign-in shortcut.
- & Unconfigured Google Apps Script webhook -> Graceful fallback: promo code is saved locally with "not_configured" delivery status so the user still receives their discount.

Workflow 2: Customer Authentication & Session Recovery

STATUS: IMPLEMENTED

Purpose: Authenticates registered customers via email, customer ID, or phone number, verifies salted password credentials, executes self-healing for demo accounts, and restores user session.



Workflow Summary & Key Components:

Execution Flow:

- Customer enters identifier and password on CustomerLoginPage (or uses Quick Demo button).
- The component normalizes the input, matching against email, internal ID, customer ID, or phone.
- authService validates the password hash against the stored record.
- A self-healing mechanism automatically updates the stored hash if the user previously had the demo password but entered their newly chosen password.

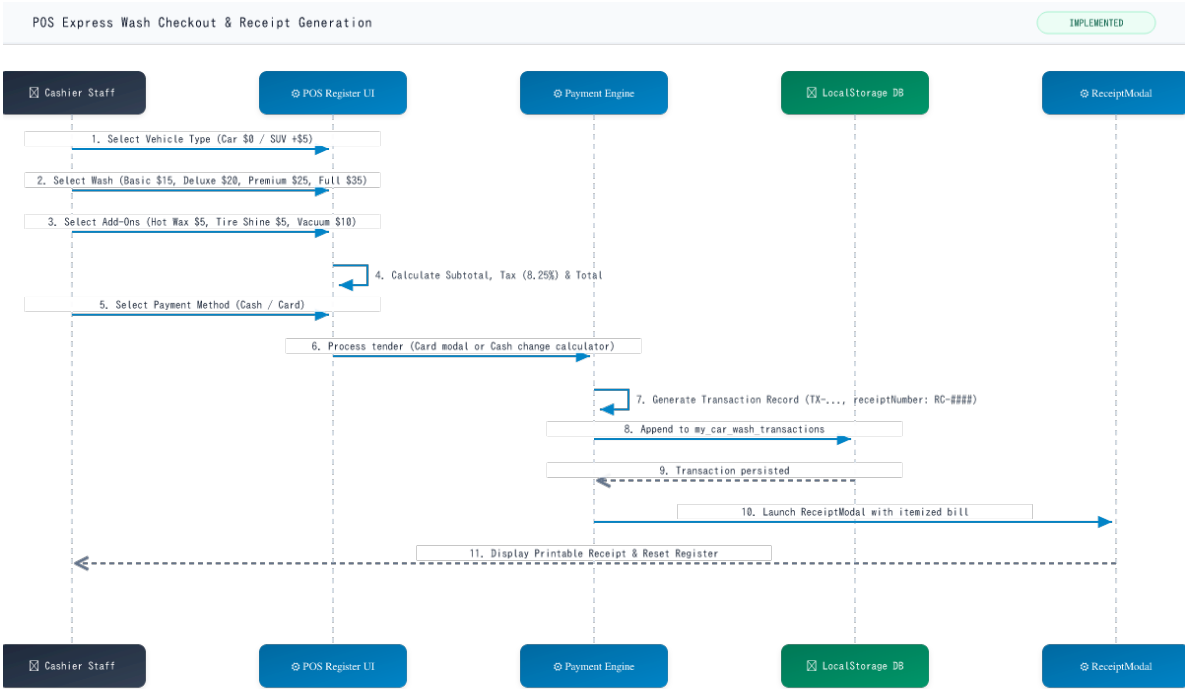
Enforced Business Rules & Error Handling:

- ' Only accounts with role = "customer" are allowed to authenticate on this portal.
- ' Invalid password or unknown account immediately displays localized error.
- ' Session persists across browser tab refreshes via localStorage.
- & Account not found -> "Invalid credentials" notification.
- & Incorrect password -> "Invalid credentials" notification with form preserved.

Workflow 3: POS Express Wash Checkout & Receipt Generation

STATUS: IMPLEMENTED

Purpose: Handles the in-lane cashier point-of-sale transaction: vehicle category selection, wash package selection, à la carte add-ons, sales tax calculation, payment processing, and printed receipt modal.



Workflow Summary & Key Components:

Execution Flow:

- Cashier selects vehicle tier (Car or SUV/Truck), wash tier, and optional add-ons in lane.
- OrderSummary computes real-time subtotal, 8.25% sales tax, and total amount due.
- Cashier selects payment method. If credit/debit card, CardPaymentPanel processes test authorization and returns masked card ref (**** 1234). If cash, change due is computed.
- App.tsx constructs a complete Transaction entity with status = "completed".

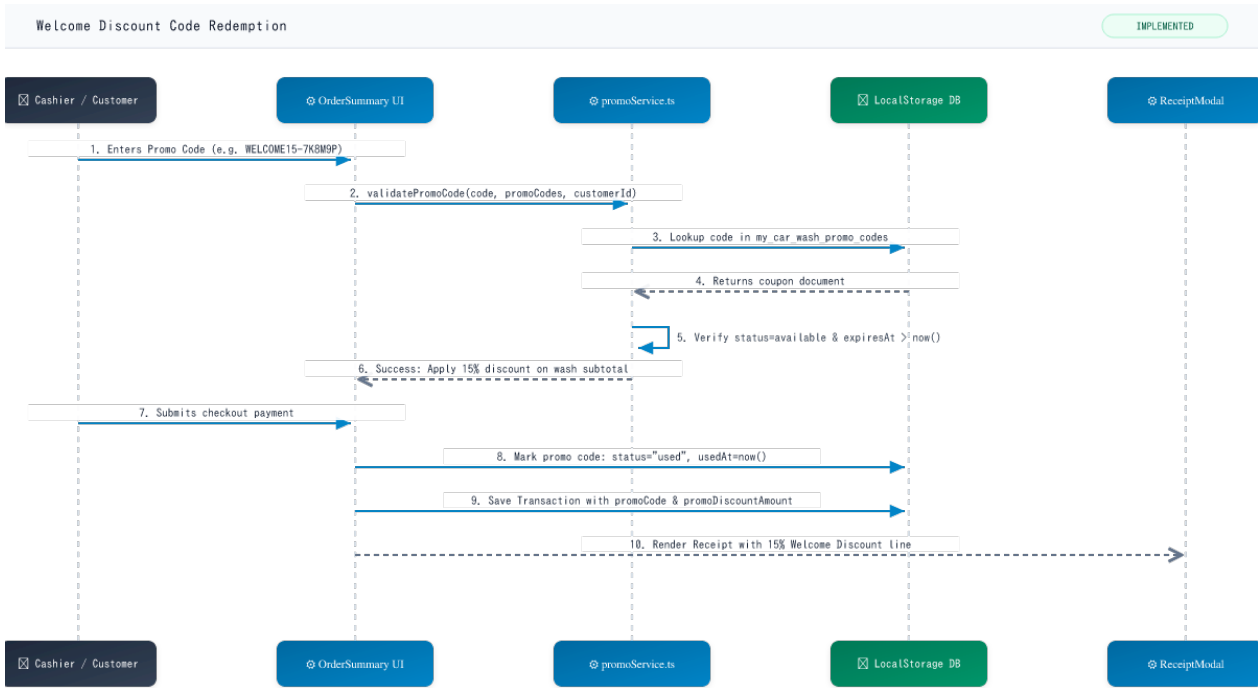
Enforced Business Rules & Error Handling:

- ' SUV/Truck category incurs configured vehicle surcharge (default +\$5.00).
- ' Sales tax is computed as subtotal * taxRate (default 8.25%).
- ' Receipt numbers are sequentially incremented for customer auditing.
- & No wash service selected -> Payment button remains disabled.
- & Cash tendered is less than order total -> Error prompt to enter full payment amount.

Workflow 4: Welcome Discount Code Redemption

STATUS: IMPLEMENTED

Purpose: Validates and applies the First-Signup 15% discount coupon at checkout, deducts discount from wash subtotal, links transaction record, and flags the coupon as redeemed.



Workflow Summary & Key Components:

Execution Flow:

- Cashier or customer enters coupon code into the OrderSummary promo input field.
- promoService checks that the code exists, belongs to eligible customer, is unused, and has not passed its 3-month expiration date.
- The cart subtotal is reduced by 15% and sales tax is recalculated.
- When transaction completes, promo code is updated to status = "used" with timestamp and receipt number.

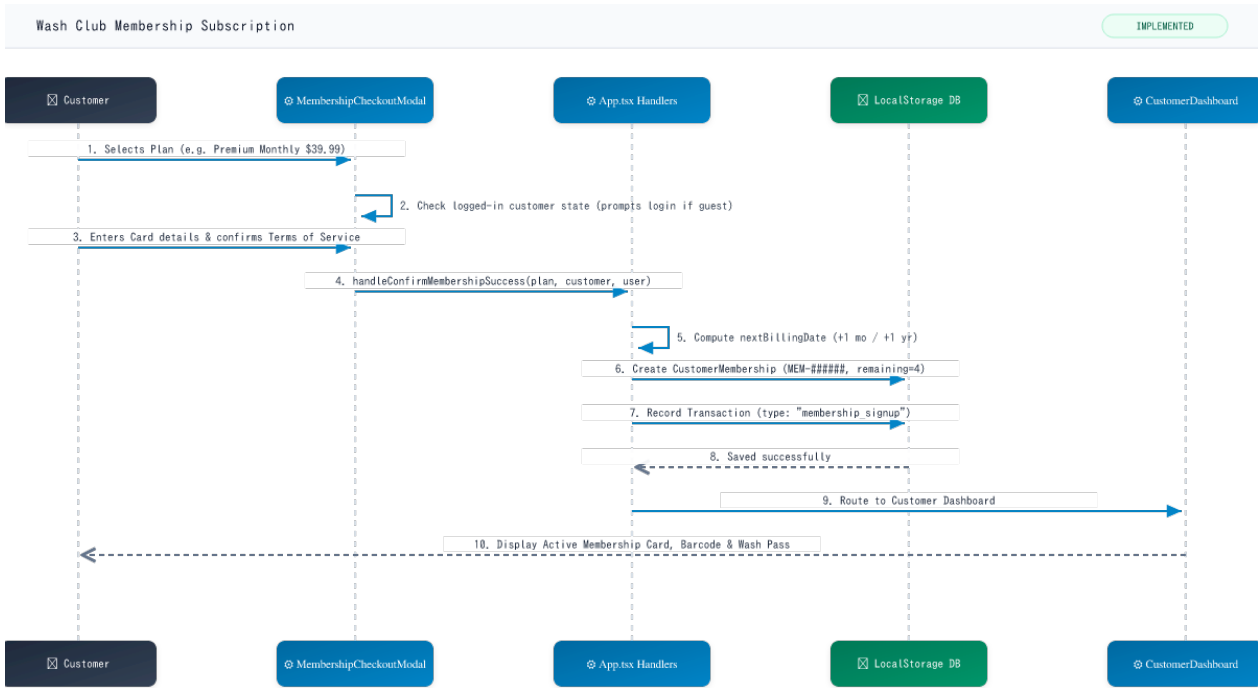
Enforced Business Rules & Error Handling:

- ' Promo code is strictly single-use; subsequent attempts are rejected.
- ' Expired codes (> 3 months from creation) cannot be redeemed.
- ' Discount applies to the wash service subtotal.
- & Non-existent code -> "Invalid promo code" message.
- & Already used code -> "This promo code has already been redeemed".

Workflow 5: Wash Club Membership Subscription

STATUS: IMPLEMENTED

Purpose: Enables customers to enroll in monthly or annual wash pass clubs, captures recurring plan terms, initiates payment, records active membership document, and registers initial wash credits.



Workflow Summary & Key Components:

Execution Flow:

- Customer navigates to Memberships page, reviews wash plans, and clicks "Join Wash Pass".
- If guest, modal prompts customer to log in or create an account. If authenticated, checkout modal opens.
- Customer enters payment method and approves subscription terms.
- System calculates billing frequency, expiration date, and allocates 4 included washes.

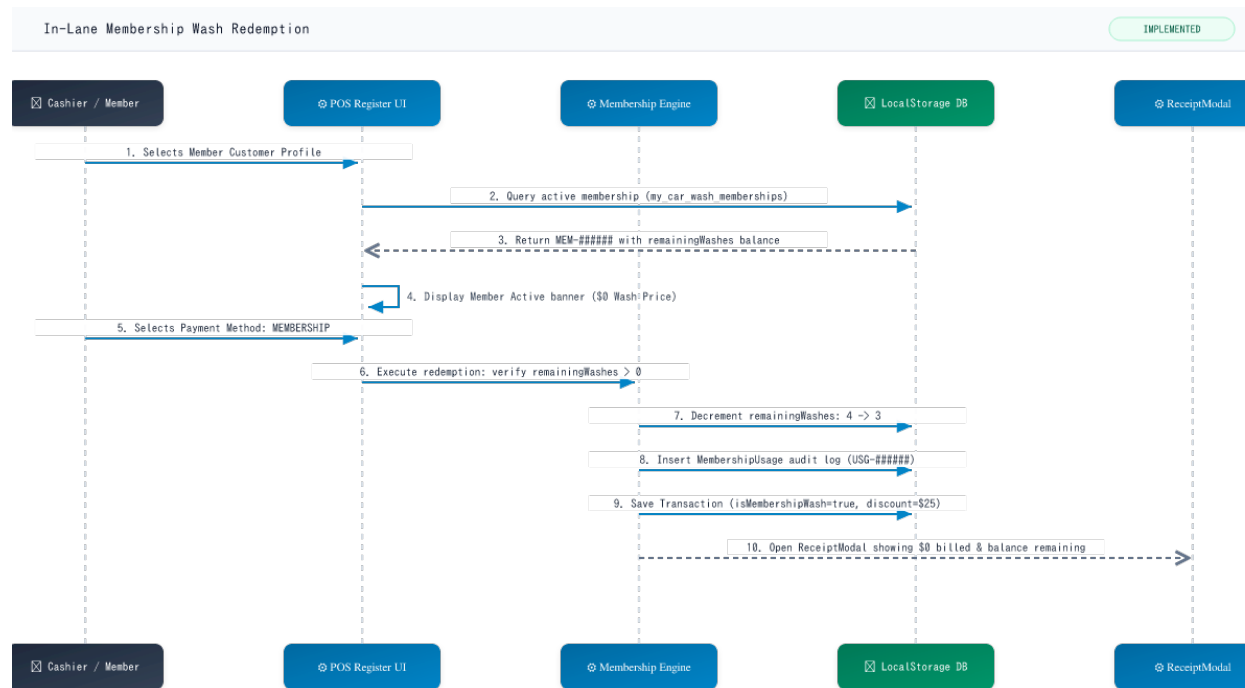
Enforced Business Rules & Error Handling:

- ' Customer can maintain an active subscription per customer ID.
- ' Initial allocation grants 4 washes for the billing period.
- ' Next billing date is computed based on billingFrequency (monthly vs yearly).
- & Unauthenticated user -> Redirects to login prompt with pre-selected plan preserved.
- & Declined card payment -> Modal displays retry message without saving membership.

Workflow 6: In-Lane Membership Wash Redemption

STATUS: IMPLEMENTED

Purpose: Identifies active club members during checkout, validates wash entitlement balance, debits 1 wash from remaining allocation, records audit log, and issues zero-dollar wash transaction.



Workflow Summary & Key Components:

Execution Flow:

- Customer arrives at wash register. Cashier selects customer profile or scans member barcode.
- System detects active subscription with positive remaining wash balance.
- The eligible plan service displays as \$0 Included With Membership.
- Cashier selects MEMBERSHIP payment method and confirms checkout.

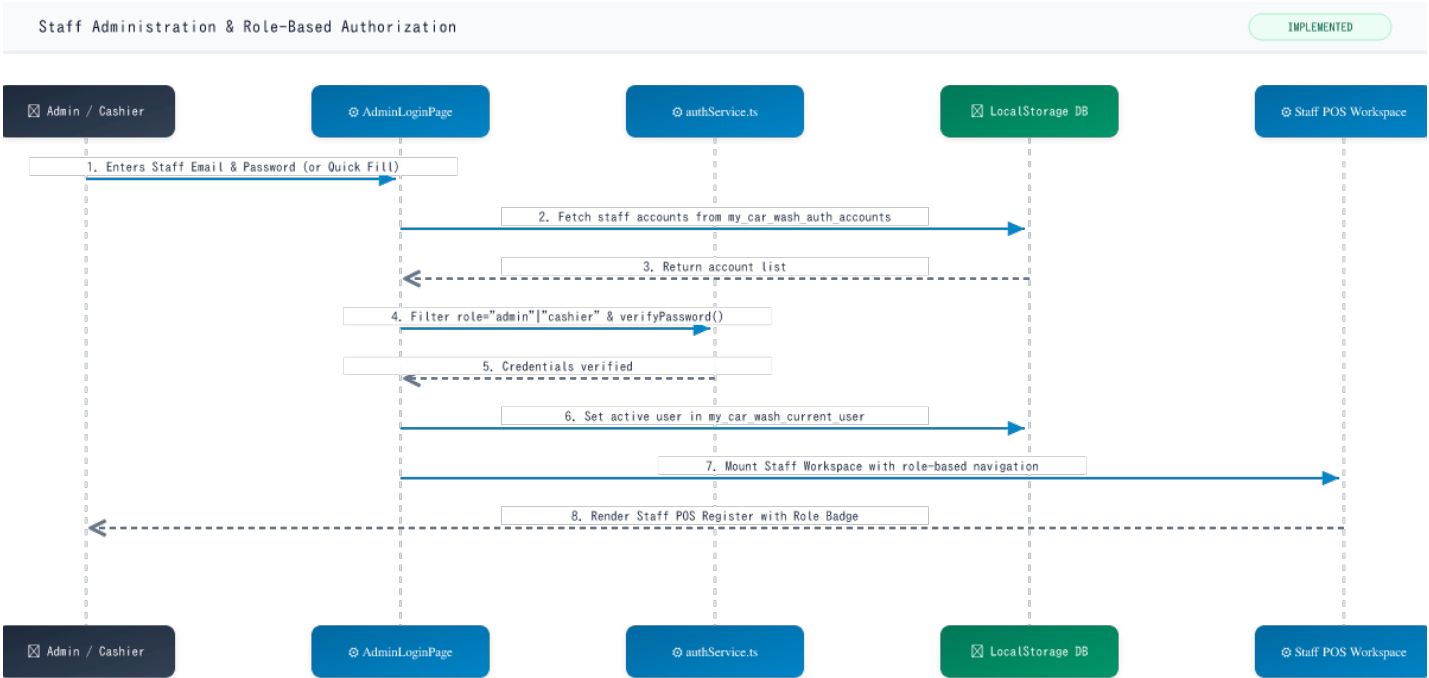
Enforced Business Rules & Error Handling:

- ' Customer must have status = "active" and remainingWashes > 0.
- ' Redeemed wash service must match plan includedServiceId.
- ' Optional add-ons are billed separately or receive plan add-on discount.
- & Zero remaining washes -> UI warns balance exhausted; prompts cashier to bill retail or renew.
- & Expired/cancelled membership -> System prevents \$0 redemption and reverts to retail pricing.

Workflow 7: Staff Administration & Role-Based Authorization

STATUS: IMPLEMENTED

Purpose: Authenticates store managers and cashier staff members, verifies password hashes, establishes staff session, and provisions role-based navigation tabs (Admin vs Cashier).



Workflow Summary & Key Components:

Execution Flow:

- Staff member navigates to AdminLoginPage and submits credentials (or clicks Admin/Cashier Quick Fill buttons).
- authService queries account directory and verifies password hash.
- User role is evaluated: "admin" receives POS, Customers, Memberships, Transactions, and Settings tabs; "cashier" receives POS, Customers, and Transactions.
- Session is persisted in localStorage.

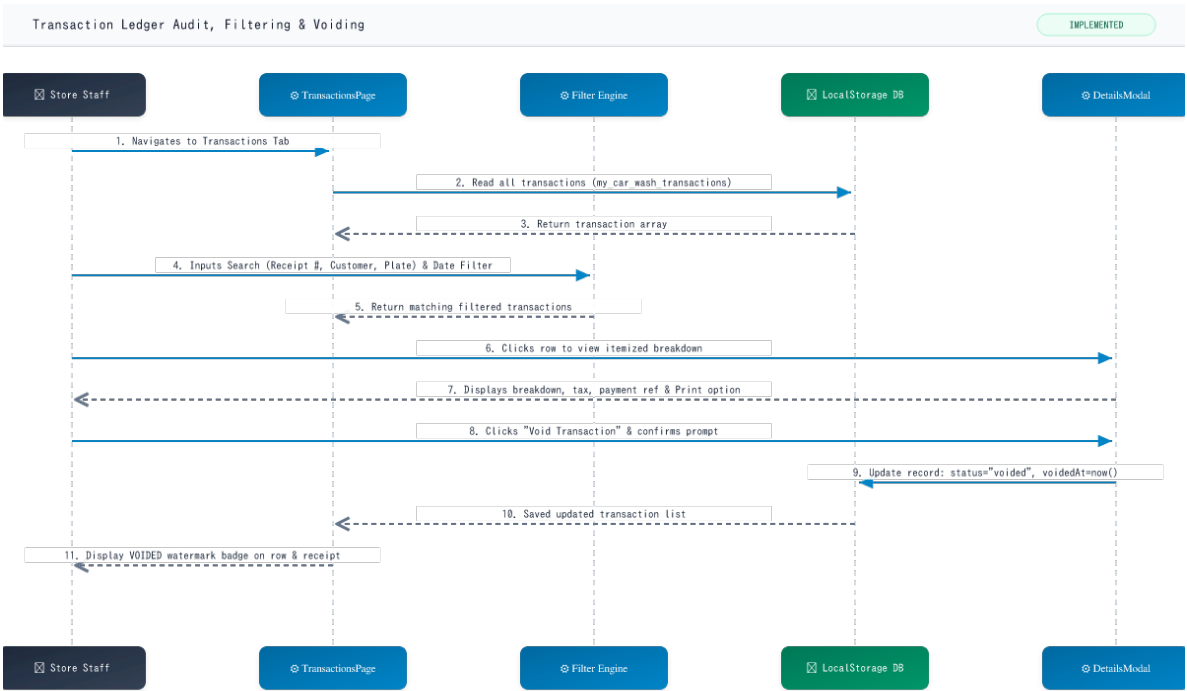
Enforced Business Rules & Error Handling:

- ' Customer accounts are blocked from accessing Admin Login.
- ' Admin role has exclusive access to modify services, vehicle surcharges, and tax rates in Settings.
- ' Password hash is verified using constant-time string comparison.
- & Invalid password -> "Invalid credentials" error alert.
- & Customer account entered on Staff Login -> "Invalid credentials" error alert.

Workflow 8: Transaction Ledger Audit, Filtering & Voiding

STATUS: IMPLEMENTED

Purpose: Provides store managers and cashiers with full transaction auditing, real-time receipt filtering, customer history lookup, receipt reprints, and transaction voiding.



Workflow Summary & Key Components:

Execution Flow:

- Staff views chronological transaction ledger displaying receipt number, timestamp, customer name, vehicle, payment method, and amount.
- Filter engine performs multi-criteria filtering by text search, status (Completed vs Voided), and payment type.
- Clicking any record opens TransactionDetailsModal with full line items, discount records, and receipt reprint option.
- Staff clicks "Void Transaction" and confirms the modal dialog.

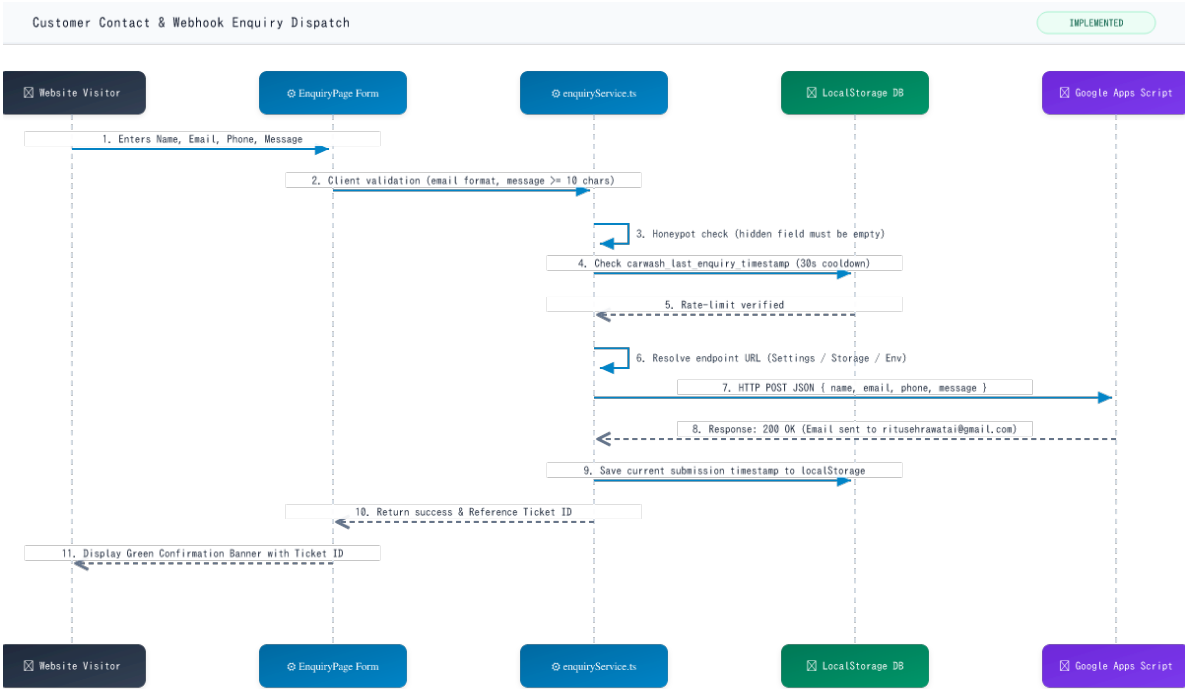
Enforced Business Rules & Error Handling:

- ' Already voided transactions cannot be voided again.
- ' Voiding retains the original financial record for audit trail integrity.
- ' Receipt modal dynamically renders "VOIDED" watermark across reprinted receipt.
- & User cancels void prompt -> Transaction remains active and completed.

Workflow 9: Customer Contact & Webhook Enquiry Dispatch

STATUS: IMPLEMENTED

Purpose: Enables website visitors to submit questions or service quotes, verifies honeypot spam protection, enforces 30-second rate limits, and forwards structured inquiries to a Google Apps Script webhook.



Workflow Summary & Key Components:

Execution Flow:

- Website visitor fills contact form on EnquiryPage with name, email, phone, and inquiry details.
- enquiryService checks honeypot field to block automated spam bots.
- A 30-second rate-limiting rule checks localStorage to prevent rapid duplicate submissions.
- The service resolves the active Google Apps Script Web App URL from Settings, localStorage, or environment variable.

Enforced Business Rules & Error Handling:

- ' Anti-spam honeypot field must remain empty.
- ' Enforces 30-second cooldown period between submissions per browser.
- ' Graceful error handling displays configuration guidance if webhook URL is unset.
- & Honeypot filled -> Silently rejected to prevent spam.
- & Submission within 30 seconds -> Displays rate-limit warning with countdown.